

Cybersecurity

Windows Active Directory Domain Services

Lesson 7.3



Learning Objectives

By the end of today's class, you will be able to:

- 1 Explain how Active Directory is used to manage enterprise-scale environments.
- 2 Define domain controllers as servers that manage AD authentication and authorization.
- 3 Use Active Directory tools to create organizational units, users, and groups.
- 4 Create and link Group Policy Objects that enforce domain-hardening policies.



Welcome



Recap

PowerShell Scripting

01

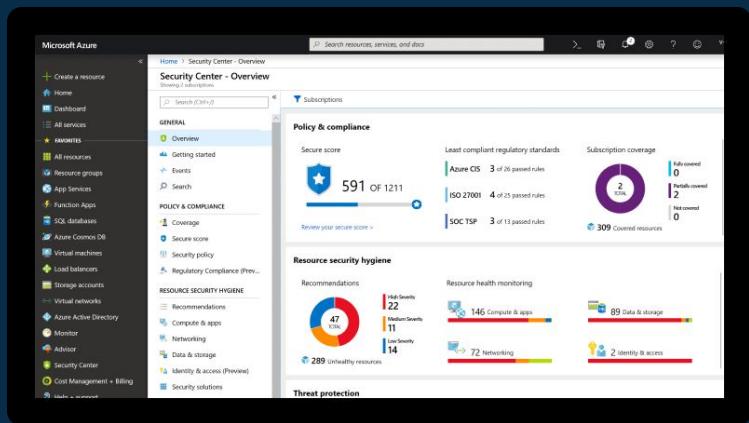
Used PowerShell to navigate system directories.

02

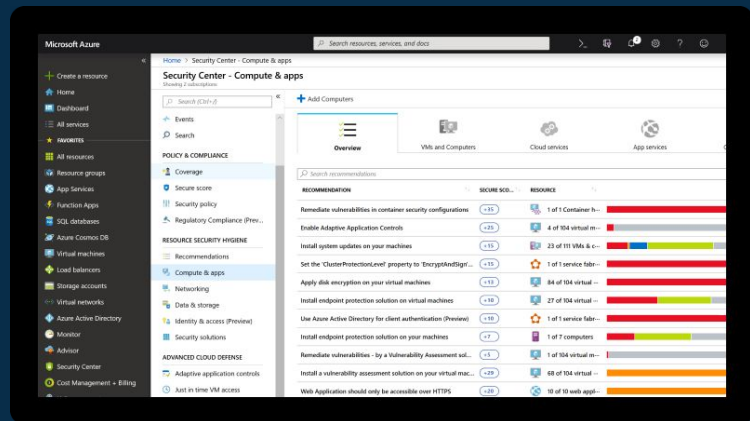
Created event logs with pipes and parameters.

03

Created script to remove Windows packages.



Today, we're going to learn how to manage Active Directory Domain Services.



Active Directory Domain Services or Active Directory (AD)



Active Directory is the central databasing and management system for enterprise-scale Windows environments.

Having a strong understanding of **Active Directory** is crucial for anyone working in Windows-based system administration and security.





Active Directory

The majority of roles in security fields interact with Active Directory.

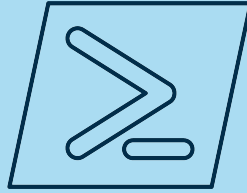
Common knowledge and understanding by:

- Security analysts
- Threat hunters
- Forensics experts
- Incident responders

Leverage vulnerabilities and exploits:

- Penetration testing
- Threat intelligence
- Malware reverse engineering





Workgroups & **Active Directory**

Workgroups & AD

Suppose a small startup with 20 employees receives a large amount of funding and adds 100 more employees to the company.

Everyone shared **resources**, but now the company needs to restrict access.

 Accountant	Needs access to	 Sensitive financial data
 IT Team	Needs access to	 Networking components like switches and routers
 Everyone	Needs access to	 Printers
 Guests	Should <i>not</i> have access to	 Network

Workgroups & AD

Security principals

are AD objects that can be authenticated, such as users and groups.

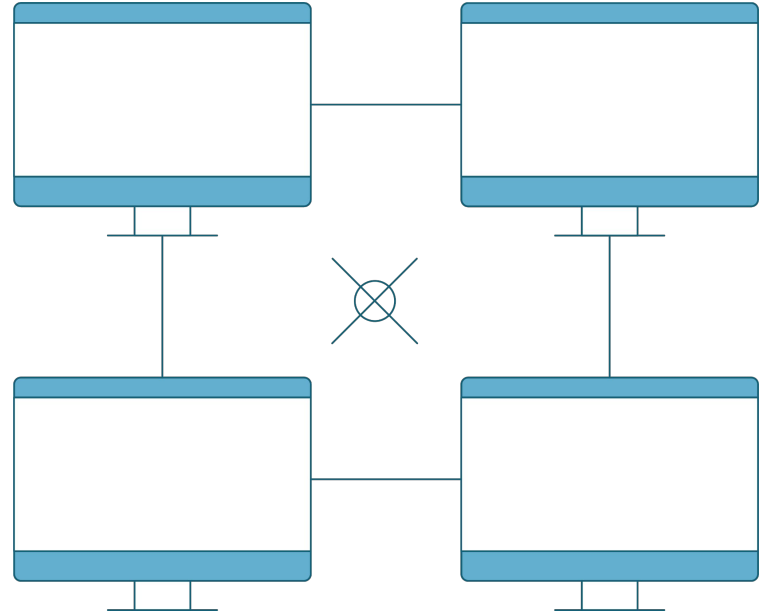
Resources are files, networking components, and printers that users need permission to access.

Security principals		Resources
 Accountant	Needs access to	 Sensitive financial data
 IT Team	Needs access to	 Networking components like switches and routers
 Everyone	Needs access to	 Printers
 Guests	Should <i>not</i> have access to	 Network

Workgroups & AD

Currently, the company is set up in a
Windows Workgroup:

- Simple setup
- Peer-to-peer
- Individual security
- Local authentication

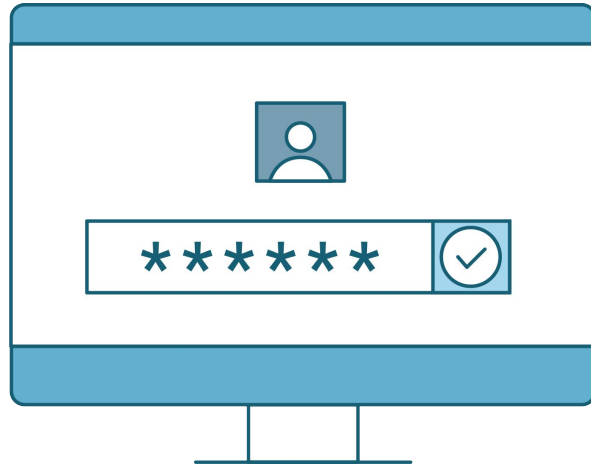




Active Directory helps us manage these resources and security principals by enforcing **authentication** and **authorization**.

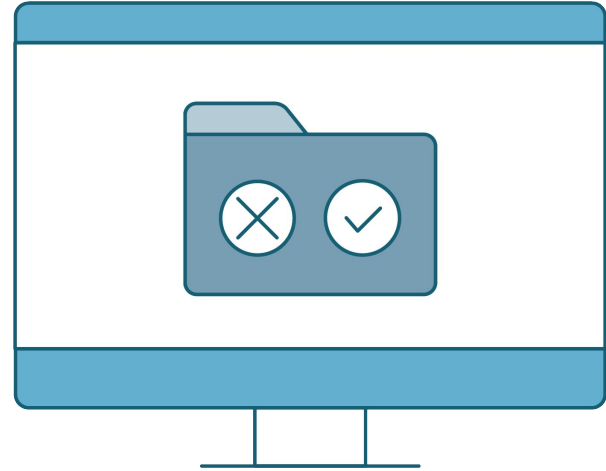
Authentication

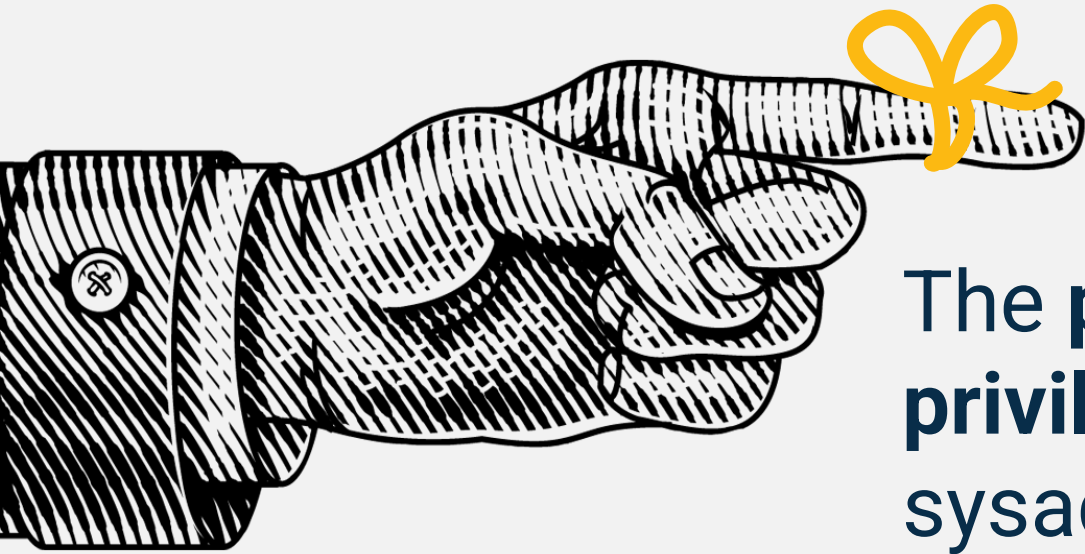
Allows users to prove their identity using a password, token, or biometric key.



Authorization

Provides or denies users permission to material.





Remember,

**The principle of least
privilege** from the Linux
sysadmin units?

Active Directory

Also helpful in the following areas:

Compliance

Implementing controls, which are crucial for passing security audits.



Risk Management

Can tie directly to business functions, such as finance, and external parties.



Access Control

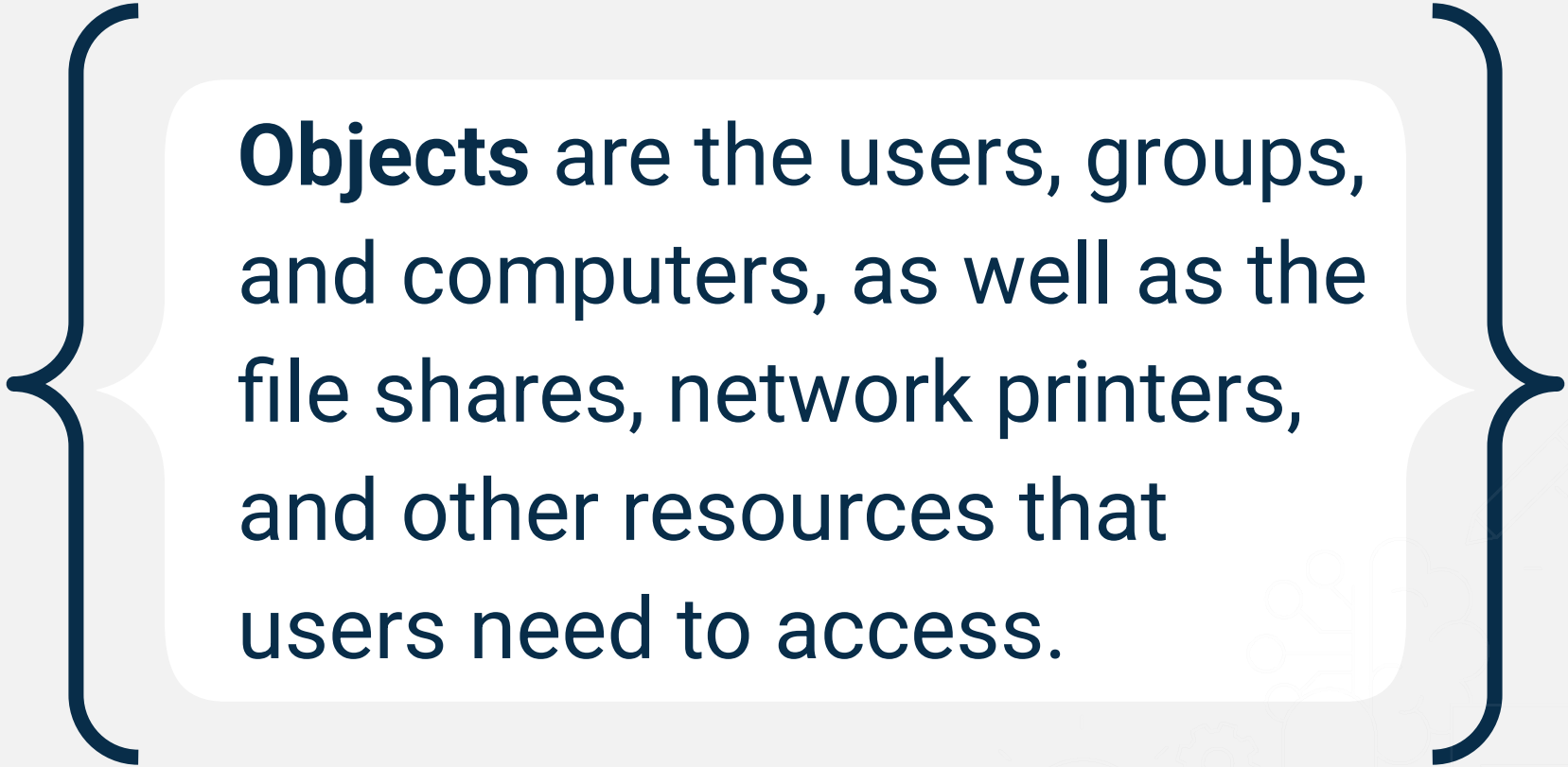
ACLs use security identifiers to manage which principals can access which resources.





Active Directory understands an organization's resources and security principals as **objects**.





Objects are the users, groups, and computers, as well as the file shares, network printers, and other resources that users need to access.



Before we can understand how AD works, let's discuss what a **Windows Server Domain** is.



A **Windows Server Domain** is a digital grouping of all object types, such as the users, computers, network resources, and security principals in an enterprise.



Windows Domains and Domain Controllers

The **Domain Controller (DC)** is a server that manages access to all the domain resources.

- Runs Active Directory
- Handles authentication
- Verifies database for requests

AD Architecture

Active Directory framework has multiple levels

Containers

Virtual collections of objects.



Organizational Units (OU)

Collections of objects that have policies applied to them.



Groups

Used to manage identical permissions and access controls for security purposes.

-Distribution Groups
-Security Groups



AD Architecture

Active Directory framework has multiple levels

Containers

Collections of one or more domains and domain trees.



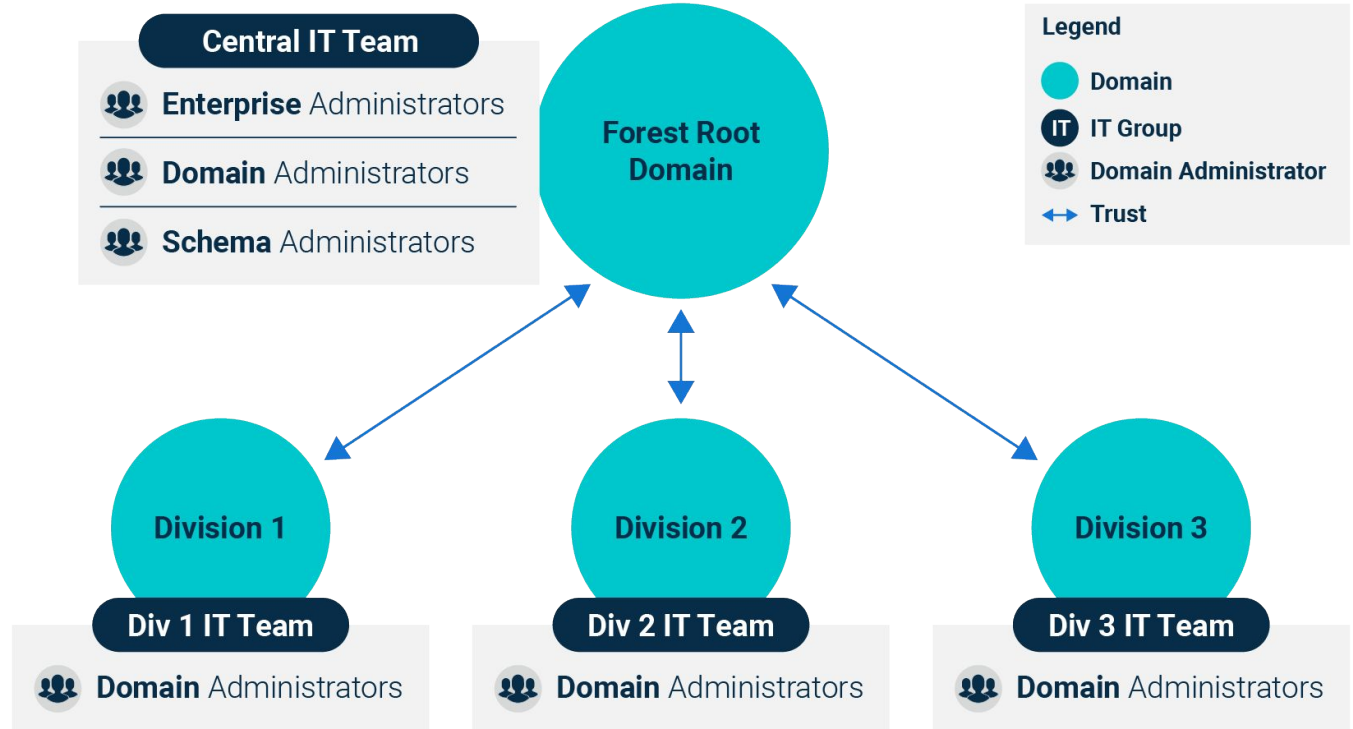
Organizational Units (OU)

Two or more trees that are part of the same AD configuration.



AD Architecture

Active Directory has a hierarchical structure of organizational units, users, machines, groups, domains, trees, and forests.



AD Authentication

AD uses the following authentication protocols:

LDAP

(Lightweight Directory Access Protocol)

A standardized protocol for adding, deleting, and editing objects.

Kerberos

A ticket-based authentication protocol, now the default authentication protocol for Windows Server domains.

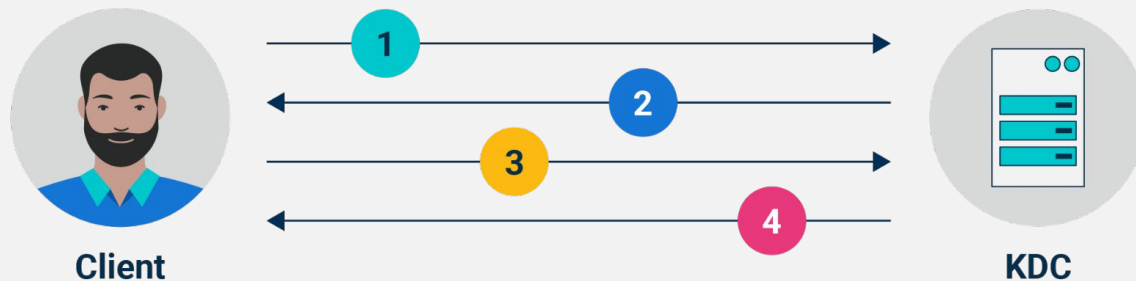
NTLM

(New Technology LAN Manager)

An authentication protocol that has become outdated because of **pass-the-hash attacks**.

Kerberos Overview

Bob is attempting to access a networked file server:



1

Bob's Windows 10 machine sends a request to authenticate the **Key Distribution Center (KDC)**, seeking a **Ticket-Granting Ticket (TGT)**. A KDC has a database of valid credentials, an **Authentication Server**, and a **Ticket-Granting Server**.

2

Once his credentials are verified, Bob receives a TGT that allows him to request access to resources. That TGT is cached and permits him to request more tickets for the current domain session.

3

When Bob attempts to access the file server, he sends the TGT to the Ticket-Granting Server, requesting access to the file server.

4

The KDC checks if the file server exists and whether the TGT is valid. If it is, the KDC sends Bob an encrypted **service ticket** containing info he authenticated earlier, and a **session key**. Bob is then sent the encrypted service ticket and a copy of the session key.

Kerberos Overview

Bob is attempting to access a networked file server:



5

Bob then uses the session key to encrypt a new message containing his current information and send that, along with the service ticket, to the file server.

6

The file server then decrypts the service ticket containing the session key, and uses that session key to decrypt the message from Bob that contains his current information.

7

Finally, the file server uses the session key to encrypt a new message to be sent to Bob containing information about the file server. If Bob's existing copy of the session key properly decrypts the message, the file server is verified.



Creating **OUs, Users, and Groups**



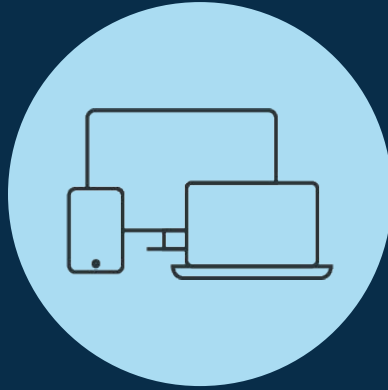
Demonstration Scenario

Creating Organizational Units

We will continue our system administration tasks and set up users in **Active Directory** so they have accounts that are logically assigned to organization units and groups.

- We need to set up marketing associate, Caroline, as a new user in Active Directory.
- To do this, we will need to **create organizational units (OUs), users, and groups.**





Instructor Demonstration

Creating Organizational Units



Questions?





Activity:

Creating Domain OUs, Users, and Groups

In this activity, you are tasked with setting up users, groups, and organizational units for your domain.

- In this scenario, your company needs you to set up the domain accounts for the people who will be using the Windows workstations.
- You will need to set up two users: **Andrew** and **Bob**. Andrew is in the development department and Bob is in the sales department.

Suggested Time:
20 Minutes



Time's up!
Let's review

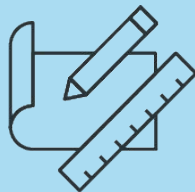


Questions?



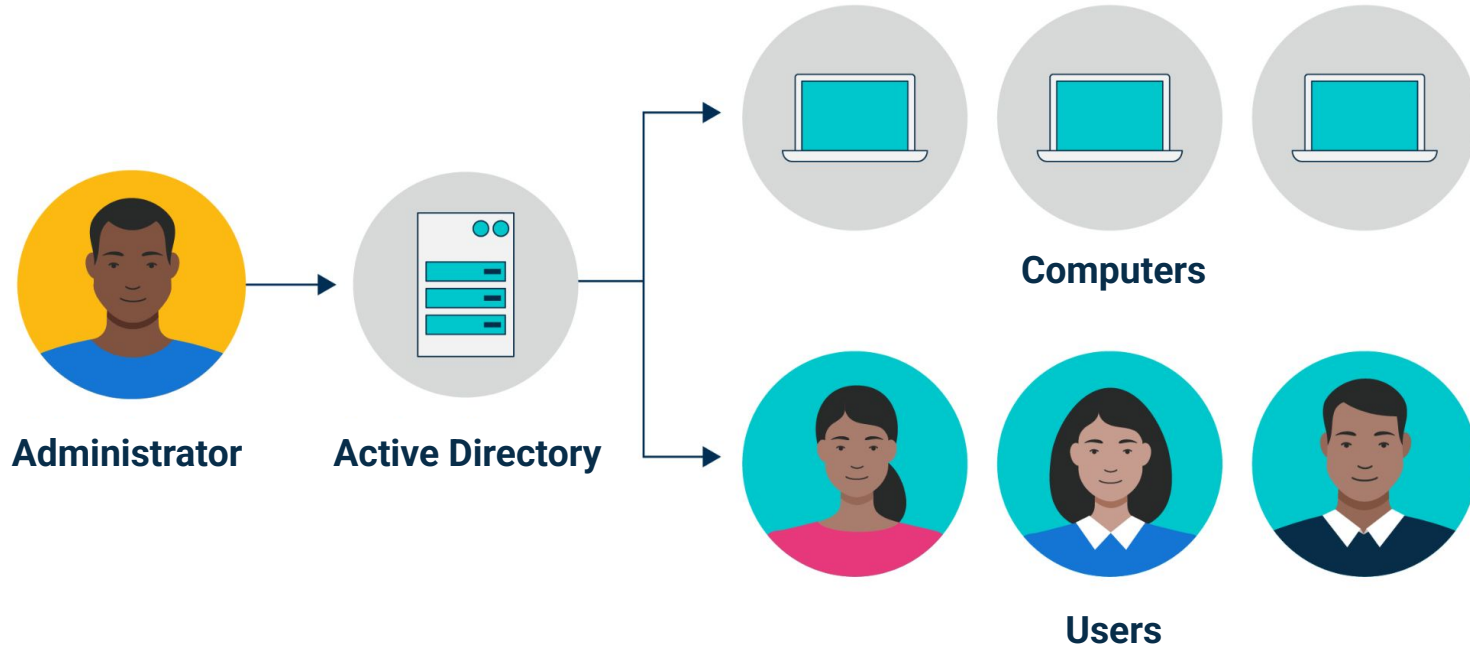


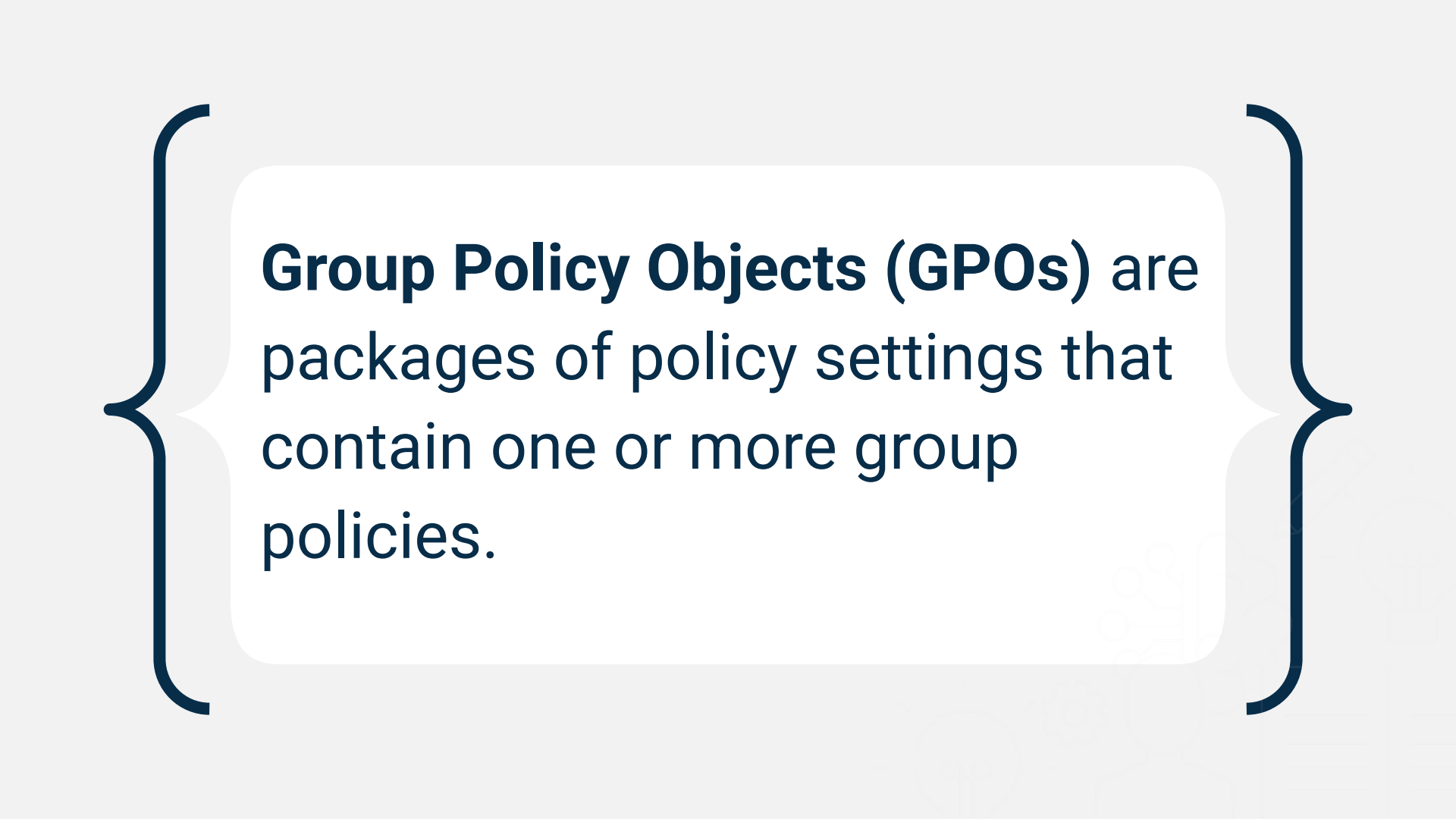
Break
15 mins



Group Policy Objects

Now that we have OUs, groups, and users, we can create Group Policies that enforce the principle of least privilege.

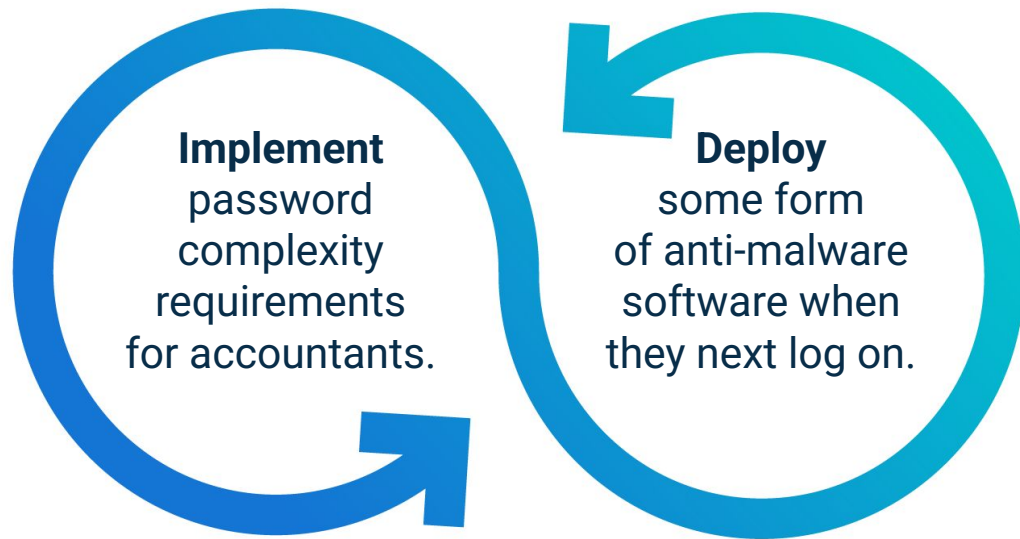




Group Policy Objects (GPOs) are packages of policy settings that contain one or more group policies.

Group Policy Objects

GPOs are the basis of AD's policy management. For example, if we want to:



We can combine these two policies into one GPO called **Better Password and Anti-Malware Setup** and apply it to all accountants in the OU.



Demonstration Scenario

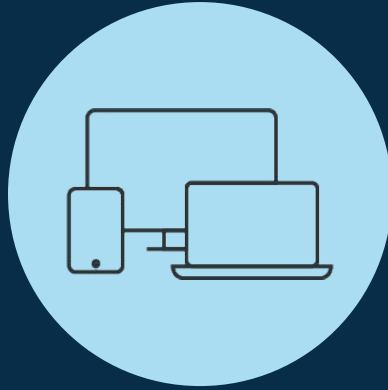
Creating Group Policy with Group Policy Objects

We want our Sales team to have limited access to Windows Control Panel.

In order to do this, we will need to:

- Create a Group Policy Object.
- Edit the individual policies for our Group Policy Object.
- Link the Group Policy Object to an organizational unit.
- Pull the latest Active Directory changes in the Windows 10 machine.
- Check if the GPO worked.





Instructor Demonstration

Create New Group Policy Object



Activity:

Creating Group Policy with Group Policy Objects

For this activity, you are continuing your role as a Windows system administrator managing Active Directory for Good Corp, Inc.

- You need to create Group Policy Objects to enforce policies for users.
- You are tasked with creating a Group Policy Object for the Sales OU that will limit the options it has in Windows Settings.

Suggested Time:
25 Minutes

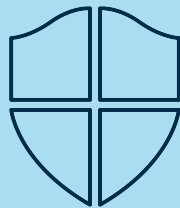


Time's up!
Let's review



Questions?



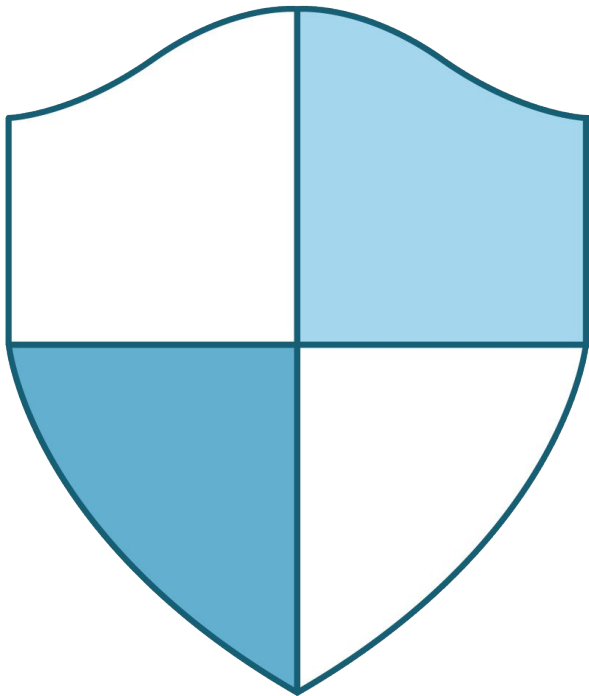


Windows **Defender**



Windows Defender

Microsoft's built-in anti-malware solution



Windows Defender can work to secure workstations via:

- Real-time protection
- Scheduled scans
- Exclusions
- Block at First Sight
- Tamper protection



Windows Defender

Key Features

01

**Cloud-delivered
protection**



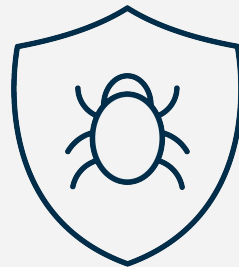
02

Family options



03

**Ransomware
protection**





Windows Defender

Key Features

04

Browser integration



05

Exploit protection



06

Offline scanning

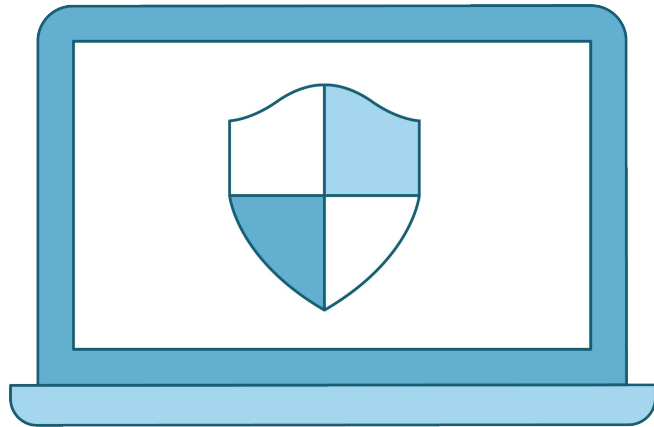


Defender Policies

Defender itself can have policies.

Policies can ensure that the security measures are uniformly applied across all workstations and servers, and they can **tailor the functionality** of Windows Defender.

- Scan preferences
- Signature updates
- Exclusions
- Firewall settings
- Sensitive folder protection



Designing **Active Directory** infrastructure is a **critical** task.

There are some **best practices** to ensure efficiency and security.



AD Best Practices

-  Fully understand the organizational structure and potential growth
-  Opt for a simpler design
-  Design the AD infrastructure to scale with growth
-  Test all GPOs in a lab or staging environment before deploying
-  Document the AD structure, policies, permissions, and any changes



Questions?





Next class

Windows Active Directory Certificate Services



Thank you